

Book exercise 14.3)

Note by (14.10) that

$$\begin{aligned} E_{AV} &= \frac{1}{M} \sum_{m=1}^M E_x [E_m(x)^2] \\ &= E_x \left[\sum_{m=1}^M \frac{1}{M} E_m(x)^2 \right] \end{aligned}$$

and from (14.11) that

$$E_{com} = E_x \left[\left\{ \sum_{m=1}^M \frac{1}{M} E_m(x) \right\}^2 \right]$$

It now follows directly from Jensen's inequality that

$$\begin{aligned} E_{com} &= E_x \left[\left\{ \sum_{m=1}^M \frac{1}{M} E_m(x) \right\}^2 \right] \\ &\leq E_x \left[\sum_{m=1}^M \frac{1}{M} E_m(x)^2 \right] \\ &= E_{AV} \end{aligned}$$

as $f(z) = z^2$ is a convex positive semidefinite function and we take the same expectation on both sides.

Book exercise 14.6

We first differentiate (14.23)

$$\begin{aligned} \frac{d}{da_m} \left\{ (e^{am/2} - e^{-am/2}) \sum_{n=1}^N u_n^{(m)} \mathbb{1}_{\{y_n(x_n) \neq t_n\}} + e^{-am/2} \sum_{n=1}^N u_n^{(m)} \right\} \\ = \frac{1}{2} \left((e^{am/2} + e^{-am/2}) \sum_{n=1}^N u_n^{(m)} \mathbb{1}_{\{y_n(x_n) \neq t_n\}} - e^{-am/2} \sum_{n=1}^N u_n^{(m)} \right) \end{aligned}$$

Setting this equal to zero and rearranging implies

$$\frac{1}{e^{am} + 1} = \frac{\sum_{n=1}^N u_n^{(m)} \mathbb{1}_{\{y_n(x_n) \neq t_n\}}}{\sum_{n=1}^N u_n^{(m)}}$$

$$\Rightarrow \frac{1}{e^{am} + 1} = \varepsilon_m$$

By (14.16)

$$\Rightarrow e^{am} = \frac{1 - \varepsilon_m}{\varepsilon_m}$$

$$\Rightarrow a_m = \ln \left(\frac{1 - \varepsilon_m}{\varepsilon_m} \right)$$

Book exercise 14.10)

Consider the optimization problem

$$\min_t \sum_{n=1}^N |t_n - t|^2$$

as described in the exercise.

We first note that $f(z) = z^2$ is a convex function. Thus (H) has a unique global minimum (if it has a minimum).

To find it, we differentiate and set equal to zero, i.e.,

$$\begin{aligned} & \frac{d}{dt} \left\{ \sum_{n=1}^N |t_n - t|^2 \right\} \\ & \quad \xrightarrow{\text{Chain rule}} = \sum_{n=1}^N 2(t_n - t) \\ & \quad = 0 \\ \Rightarrow & \quad 2 \sum_{n=1}^N t_n - 2 \sum_{n=1}^N t = 0 \\ \Rightarrow & \quad 2 \sum_{n=1}^N t_n - 2N \cdot t = 0 \\ \Rightarrow & \quad t = \underbrace{\frac{1}{N} \sum_{n=1}^N t_n}_{\text{mean of } \{t_n\}} \end{aligned}$$